

Evaluating GPT-4o for Rare Disease Diagnosis: A Stratified Benchmarking Study Across Disease Rarity Tiers

Oluwaseyi Omisore

University of Massachusetts Boston

2026

Abstract

Rare diseases affect over 300 million people worldwide, yet patients wait an average of 4 to 7 years for a correct diagnosis. This study asks: Can a large language model (LLM) help? I benchmarked GPT-4o on 296 clinical case summaries spanning six disease rarity tiers. Surprisingly, accuracy did not decrease as diseases got rarer. Instead, I found a non-monotonic pattern peaking in the middle tiers: Very rare diseases (2-5 training examples) achieved 37.5% top-1 accuracy while Ultra-rare diseases (1 training example) achieved only 11.4%. For the Rare tier, the correct diagnosis appeared in the model's top-5 list 50% of the time, yet was ranked first only 33.9% of the time, suggesting a confidence problem rather than a knowledge problem. These findings are statistically significant ($p=0.011$). These results show that LLMs are not equally useful across the rare disease spectrum and that different rarity tiers require different improvement strategies.

Keywords: *rare disease diagnosis, large language models, GPT-4o, LLM benchmarking, diagnostic accuracy, PACE framework*

1. Introduction

Rare diseases are defined as conditions affecting fewer than 1 in 2,000 people. While each disease is individually uncommon, there are over 7,000 known rare diseases, and together they affect 300 million people worldwide. Despite this scale, patients with rare diseases face a difficult reality: the average time from first symptom to correct diagnosis is 4 to 7 years. During this time, patients often see multiple specialists, receive incorrect treatments, and experience preventable complications.

The core problem is a mismatch between how rare diseases are distributed and how medical knowledge is built. Clinicians learn by seeing patients, but a doctor might see only one or two cases of a truly rare disease in their entire career. This makes pattern recognition, the foundation of clinical diagnosis, very difficult for the rarest conditions.

Large language models (LLMs) like GPT-4o have been trained on enormous amounts of medical text, including case reports, textbooks, and clinical literature. In theory, an LLM could have read every published case of every rare disease, giving it a breadth of exposure no individual clinician could match. This raises a practical question: how well can an LLM actually diagnose rare diseases from clinical descriptions, and does its performance depend on how rare the disease is?

This project addresses that question through a structured benchmarking study, following the PACE data science framework (Plan, Analyze, Construct, Execute). I evaluated GPT-4o on 296 clinical case summaries across six disease rarity tiers and report accuracy, error patterns, and practical implications for when and how LLMs might be useful in rare disease workflows.

2. Data

2.1 Dataset Overview

I used the publicly available Kaggle dataset "Evaluate LLMs for Rare Disease Diagnosis." The training set contains 50,594 clinical case summaries covering 1,393 unique rare diseases. The evaluation set contains 6,915 cases covering 1,012 diseases. Each case includes a free-text clinical description written by a physician, the correct disease name, known synonyms, and medical coding identifiers (ICD-10, ICD-11, OMIM, GARD, UMLS).

2.2 Data Quality Notes

Medical ontology identifiers were missing for a significant portion of cases: OMIM IDs were absent for about 35% of cases, and GARD IDs for about 21%. These were kept in the dataset since they are reference tags, not features used in benchmarking. Case summary lengths varied widely, from under 100 to over 60,000 characters, with a median of roughly 1,900 characters. No duplicate case IDs were found.

2.3 Known Limitations of the Dataset

The dataset skews toward Congenital and Chromosomal diseases (about 19% of training cases) and Neoplasms (about 16%). This means results may be more reliable for these categories than for rarer specialties like metabolic or neurological diseases. Additionally, the disease frequency distribution follows a long tail: 158 diseases appear only once in training, while the most common disease appears over 800 times. This imbalance is the core variable I am studying for this project.

3. Methods

3.1 PACE Framework

This project followed the PACE framework throughout. In the Plan phase, I defined my research question, rarity tier boundaries, and success metrics before touching the data. In the Analyze phase, I audited data quality and documented missing values and biases. In the Construct phase, I built the benchmarking pipeline and matching system. In the Execute phase, I ran the benchmark, generated figures, and interpreted results with clinical context.

3.2 Defining Rarity Tiers

I defined rarity operationally: a disease's rarity tier is determined by how many times it appears in the training set. This reflects how much training signal the model had for each disease. Six tiers were created:

Tier	Training Freq.	What it means
Zero-shot (novel)	0	Disease never seen in training
Ultra-rare (1)	1	The model has seen this disease exactly once
Very rare (2-5)	2-5	Very limited exposure (a few case descriptions)
Rare (6-20)	6-20	Moderate exposure (enough to recognize patterns)
Moderate (21-100)	21-100	Substantial training exposure
Common (100+)	>100	High exposure within the rare disease context

3.3 Sampling

I used stratified random sampling to draw cases equally from each rarity tier rather than randomly from the full dataset. This ensures every tier is fairly represented. Without stratification, common tiers would dominate, and rare tiers would have too few cases to analyze. My target was 50 cases per tier using random seed 42 for reproducibility. Final sample sizes ranged from 28 (Zero-shot tier, limited by dataset availability) to 56 per tier, totaling 296 cases.

3.4 Model and Prompting

I used GPT-4o accessed through GitHub Models at zero cost. The temperature was set to 0, so the model gives consistent responses. I used two prompt types for each case. The top-1 prompt asked for the single best diagnosis in JSON format. The top-5 prompt asked for five ranked diagnoses. Case summaries were truncated to 3,000 characters. Most cases fit comfortably within this limit. Only cases in the longest 30% of the distribution were affected by truncation. Truncation was applied consistently at the end of the text, so earlier clinical details were preserved. Future work should test whether truncation affects accuracy for longer cases.

3.5 Matching Strategy

LLMs rarely reproduce disease names with exact spelling and capitalization. "Cystic Fibrosis" and "cystic fibrosis" should both count as correct. I used a three-level matching system: exact match after lowercasing and removing punctuation; synonym match against known alternative names in the dataset; and fuzzy match using word overlap (Jaccard similarity threshold 0.6). A prediction only counts as wrong if it fails all three

levels. This approach makes accuracy estimates more fair and less sensitive to naming conventions.

3.6 Baseline Comparison

To ground my results, I calculated a random chance baseline for each tier: if a model guessed randomly from all diseases in the evaluation set (1,012 diseases), it would be correct approximately 0.1% of the time. Even the lowest-performing tier (Ultra-rare at 11.4%) dramatically exceeds this baseline. This confirms the model has genuine medical knowledge. However, 11.4% remains far below clinical utility, which requires much higher precision for safety-critical applications.

4. Results

4.1 Overall Performance

Across all 296 cases, GPT-4o achieved an overall top-1 accuracy of 25.7% (76 correct out of 296) and top-5 accuracy of 31.8%. While this clearly exceeds random chance (0.1%), it also falls well short of clinical reliability. The results become more meaningful when broken down by rarity tier.

4.2 Accuracy by Rarity Tier

Table 1 and Figure 1 show the complete results. The key finding is that accuracy does not decrease smoothly as diseases get rarer. Instead, it follows a non-monotonic curve, peaking in the middle tiers.

Table 1. GPT-4o Diagnostic Accuracy by Disease Rarity Tier

Rarity Tier	N	Top-1 Acc	95% CI	Top-5 Acc	Gap (pp)
Zero-shot (novel)	28	10.7%	[3.7%, 27.2%]	14.3%	3.6
Ultra-rare (1)	44	11.4%	[5.0%, 24.0%]	11.4%	0.0
Very rare (2-5)	56	37.5%	[26.0%, 50.6%]	42.9%	5.4
Rare (6-20)	56	33.9%	[22.9%, 47.0%]	50.0%	16.1
Moderate (21-100)	56	21.4%	[12.7%, 33.8%]	25.0%	3.6
Common (100+)	56	28.6%	[18.4%, 41.5%]	33.9%	5.4

Gap (pp) = Top-5 minus Top-1 accuracy. A higher gap means the model has the right answer but ranks it too low.

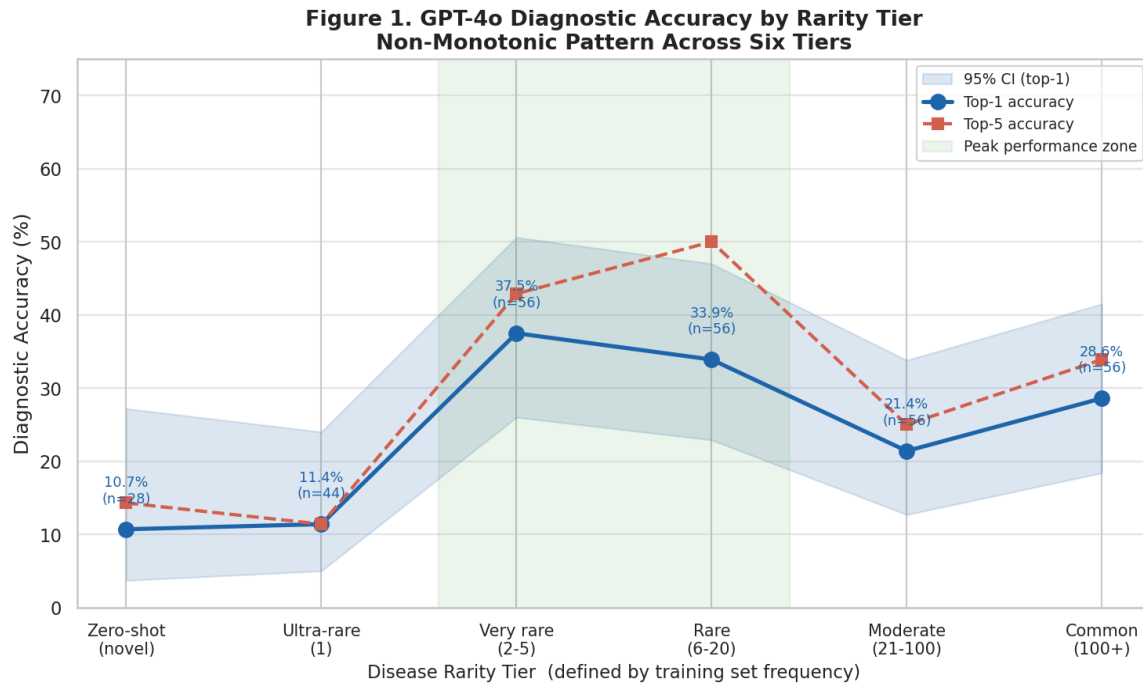


Figure 1. GPT-4o accuracy by rarity tier. The blue line (top-1) and dashed red line (top-5) both show a non-monotonic pattern, peaking in the middle tiers. The shaded blue region shows 95% confidence intervals. The green band marks the peak performance zone.

4.3 Statistical Significance

To confirm this pattern is not due to chance, I ran a chi-square test across all six tiers ($\chi^2=14.88$, $df=5$, $p=0.011$). This means there is only a 1.1% probability of seeing this pattern by random chance. It confirms that rarity tier is a statistically significant predictor of GPT-4o accuracy. A pairwise Fisher exact test comparing Ultra-rare and Very rare gave $p=0.005$, confirming the most dramatic contrast is robust even with smaller sample sizes.

4.4 Two Different Types of Failure

Figure 2 reveals something important: not all wrong answers are the same. The gap between top-1 and top-5 accuracy (Gap column in Table 1) tells us why the model was wrong.

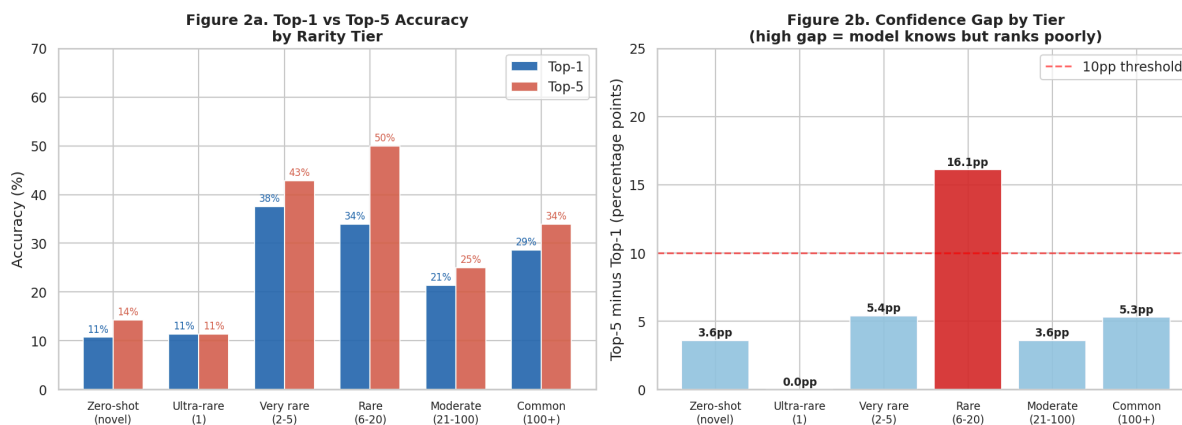


Figure 2a (left) shows top-1 vs top-5 accuracy side by side. Figure 2b (right) shows the gap between them. Red bars exceed the 10 percentage point threshold, indicating a ranking failure rather than a knowledge failure.

For the Rare tier (6-20 training examples), the gap is 16.1 percentage points. The correct answer appears in the model's top-5 list 50% of the time, but is ranked first only 33.9% of the time. The model has the knowledge but lacks the confidence to rank it first. This is a calibration failure.

For the Ultra-rare tier, the gap is exactly 0.0 percentage points. Top-1 and top-5 accuracy are both 11.4%. When the model is wrong, the correct answer is not anywhere in its top 5. The model simply does not know, so this is a knowledge failure.

These two failure modes require completely different solutions, which have direct implications for future work.

4.5 Where Does the Correct Answer Appear?

Figure 3 shows the rank position of correct diagnoses across tiers, breaking down exactly where in the model's ranked output the correct answer lands.

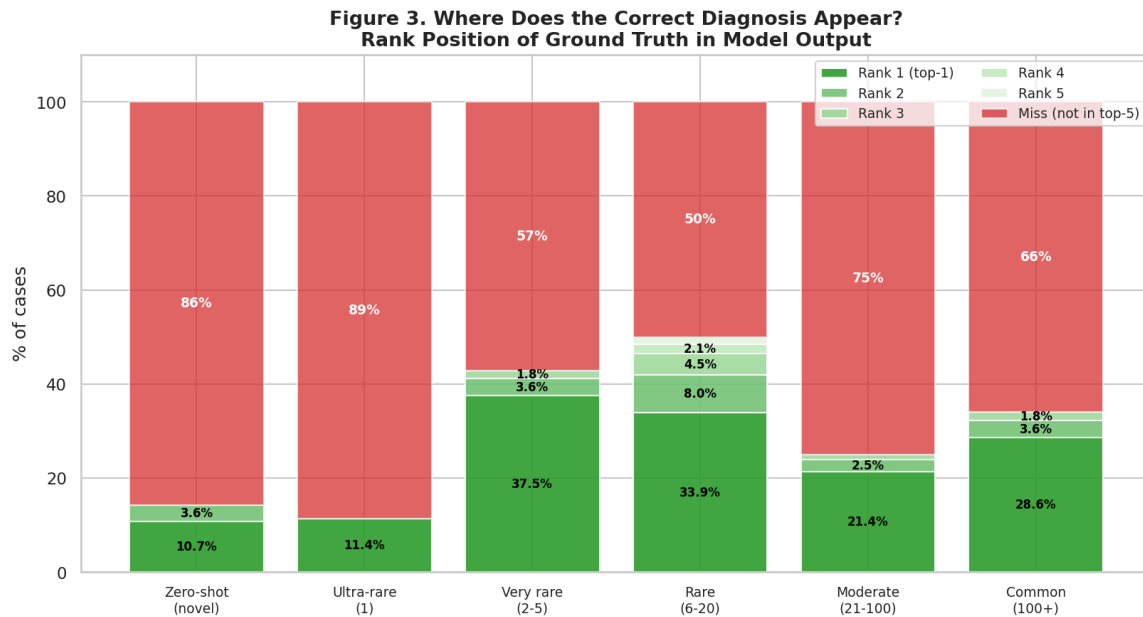


Figure 3. Rank position of the correct diagnosis in the model's top-5 output. Green shades indicate correct answers at different ranks. Red indicates the correct answer was not in the top 5 at all. The Rare tier has the most cases where the correct answer appears at rank 2 or lower, consistent with a calibration failure rather than a knowledge gap.

4.6 Error Analysis

Among 220 incorrect predictions, I observed meaningful patterns beyond individual wrong answers. The most frequent wrong prediction was Angelman syndrome (3 occurrences). Several other syndromes appeared twice as wrong answers: Silver-Russell syndrome, Cornelia de Lange syndrome, Ellis-van Creveld syndrome, and Beckwith-Wiedemann syndrome.

These errors are clinically interesting. Angelman syndrome, Silver-Russell syndrome, and Beckwith-Wiedemann syndrome are all imprinting disorders. These are conditions caused by abnormal expression of genes that differ depending on which parent contributed them. The model appears to confuse diseases within this mechanistic category, suggesting it may be pattern-matching on surface-level symptom overlap rather than the underlying genetic mechanism. Similarly, Cornelia de Lange and Ellis-van Creveld are both chromosomal syndromes with overlapping features like growth restriction and limb abnormalities.

Two cases produced the response "Insufficient clinical information for diagnosis. " This was a medically appropriate acknowledgment of uncertainty rather than a hallucinated answer. While unhelpful for benchmarking, this behavior is preferable to a confident wrong answer in a clinical setting.

5. Discussion

5.1 Why Does Accuracy Peak in the Middle?

My central finding that accuracy peaks in the Very rare and Rare tiers rather than in more common disease categories challenges the simple assumption that more training data always leads to better performance. I propose that diseases in these middle tiers benefit from a specific combination of factors: they are rare enough to have distinctive, recognizable clinical presentations, but common enough to appear in medical literature with sufficient detail for an LLM to learn their key features.

Ultra-rare diseases, by contrast, may have been described only once or twice in all of medical literature. A model trained on that limited signal simply cannot build reliable recognition. Common rare diseases may have more variable presentations, making any single case harder to classify correctly despite abundant training examples.

5.2 When Should a Clinician Trust This Model?

Based on my findings, I can offer specific guidance. For diseases in the Very rare and Rare tiers, GPT-4o may be most useful as a differential diagnosis generator. The 50% top-5 accuracy in the Rare tier means that in half of the cases, the correct diagnosis appears somewhere in the model's five suggestions. A clinician who uses these suggestions as a starting point, rather than a final answer, could benefit meaningfully.

For Ultra-rare diseases, GPT-4o should not be relied upon without additional support. With a top-5 accuracy of only 11.4%, the model fails to surface the correct diagnosis in nearly 9 out of 10 cases. These are precisely the diseases where diagnostic delay is most harmful and where retrieval-augmented approaches would be most valuable.

6. Ethical Considerations

6.1 Patient Harm Risk

A wrong diagnosis in a rare disease context can mean years of treatment for the wrong condition while the actual disease progresses. The stakes are highest for the tiers where our model performs worst. Any LLM-based diagnostic tool must be framed explicitly as decision support, not as a replacement for clinical judgment, and must include confidence thresholds below which the system declines to make recommendations.

6.2 Bias in the Data and Results

This dataset over-represents Congenital and Neoplasm disease categories. Results in these domains may be more reliable than for Metabolic or Neurological rare diseases. Additionally, my matching strategy may still undercount correct diagnoses when models use valid but unusual terminology. The true accuracy may be slightly higher than reported.

6.3 Data Contamination

GPT-4o was trained before our benchmarking date, and some of the clinical case summaries in this dataset may have appeared in its training data. This means my accuracy estimates could be inflated relative to true performance on genuinely new cases. Future benchmarks should use collected cases unavailable on the public internet.

6.4 Responsible Deployment

Following responsible AI principles for clinical contexts, any deployment recommendation derived from this research must include: human oversight at every diagnostic decision point; defined escalation pathways when model confidence is low; transparency to patients about the role of AI in their care; and ongoing monitoring of real-world performance. This study establishes a quantitative baseline against which future deployments can be measured.

6.5 Synthesizing What the Data Tells Us

A 37.5% top-1 accuracy in the Very rare tier means the model is wrong nearly two-thirds of the time, even at its best. These results should motivate investment in knowledge-grounded approaches and careful human oversight, not premature deployment of current models as standalone diagnostic tools.

7. What the Analysis Found vs. What I Observed

7.1 What the Analysis Found

- A statistically significant non-monotonic accuracy pattern across rarity tiers (chi-square $p=0.011$)
- Ultra-rare diseases had a 0.0pp top-1 to top-5 gap, indicating complete knowledge failure.
- Rare diseases had a 16.1pp gap, indicating a calibration failure.
- Overall top-1 accuracy of 25.7% across 296 cases covering 257 unique diseases.
- Error clustering around imprinting disorders and chromosomal syndromes.
- Two cases produced appropriate uncertainty responses rather than wrong diagnoses.
- All results exceeded the random baseline of approximately 0.1% by a wide margin.

7.2 What I Observed

- I learned that growth doesn't always have to be large to be measurable. In the context of rare diseases, little incremental progress in diagnostic accuracy is a significant step toward ending a patient's years of uncertainty.

- I initially expected a one-to-one relationship between diseases and names, but discovered that the same condition often has multiple synonyms, which made the data much more complex than I anticipated.
- If I were to repeat this project, I would collaborate with a biology or pre-med major to gain deeper clinical insight into the "near-miss" errors and better understand the underlying genetic mechanisms.

8. Project Cost and Reproducibility

8.1 Cost Breakdown

This entire project was completed at zero direct monetary cost using the GitHub Models free tier, which allows 50 API requests per day. Each of the 296 benchmarked cases required 2 API calls (one top-1, one top-5), totaling 592 requests collected over approximately 12 days of daily runs.

Resource	This Pilot Study	Full Scale (6,915 cases)
API calls required	592	~13,830
GitHub Models (free)	Fully covered — \$0	Not feasible (~277 days)
Anthropic Claude API	~\$0.30 estimated	~\$7-15 estimated
UMass Boston Gibbs HPC	Not used (pilot only)	Free with faculty sponsor
Total direct cost	\$0.00	\$0 with HPC access

8.2 Time Investment

- Dataset setup and environment: 1 day
- PACE data cleaning and documentation: 1 day
- Daily benchmark runs: 12 days elapsed, ~30 minutes active work per day
- Analysis and figure generation: 1 day
- Report writing: 2 days
- Total: approximately 17 days elapsed, roughly 5-6 days of active work

8.3 Reproducibility

All code is documented across four Jupyter notebooks available at github.com/sharlerx. Random seed 42 is used throughout. Model: GPT-4o via GitHub Models. Benchmarking period: April-May 2025. The checkpoint system ensures no cases are duplicated, and all results are recoverable. Anyone with a free GitHub account can reproduce this pilot study at zero cost.

9. Limitations

- Pilot scale: n=28-56 per tier. The full evaluation set has 6,915 cases. Results are directional and should be confirmed at full scale.
- Single model: Only GPT-4o was tested. Claude, Llama, and Gemini may perform differently.
- Zero-shot prompting only: I did not test retrieval-augmented generation, chain-of-thought prompting, or fine-tuned models.
- Matching under-counting: My three-level matching system may still miss correct diagnoses using unusual but valid terminology.
- Dataset bias: Over-representation of Congenital and Neoplasm categories limits generalizability across all specialties.
- Possible data contamination: GPT-4o may have seen some of these cases during training, potentially inflating accuracy estimates.
- Truncation effect: Cases longer than 3,000 characters were cut off. The impact of truncation on accuracy was not directly measured.

10. Future Work

- Full-scale evaluation: Run all 6,915 evaluation cases using UMass Boston Gibbs HPC cluster with open-source models at no API cost.
- Multi-model comparison: Benchmark Claude, Llama 3, and Gemini under identical conditions to identify model-specific strengths by rarity tier.
- Retrieval-augmented generation: Add OMIM and Orphanet database context to prompts for Ultra-rare cases and measure accuracy improvement.
- Rarity-aware re-ranking: Train a lightweight model that takes a rarity tier as input and re-ranks LLM differential diagnoses to target the calibration failure observed in the Rare tier.
- Adaptive prompting: Automatically adjust the prompt strategy based on predicted rarity so it has more context-rich prompts for rarer diseases and simpler prompts for common ones.
- ICD domain breakdown: Analyze accuracy separately for Congenital, Metabolic, and Neurological disease categories to identify specialty-specific failure modes.

11. Conclusion

This project presents a structured pilot benchmarking study of GPT-4o on rare disease diagnosis, conducted at zero cost using freely available tools and the PACE data science framework. The central finding, a non-monotonic accuracy pattern peaking in the Very rare and Rare tiers, challenges the intuitive expectation that more training data

always means better performance. More importantly, it reveals two distinct failure modes: a knowledge gap for Ultra-rare diseases (where the correct answer is never in the top 5) and a confidence calibration problem for Rare diseases (where the correct answer is in the top 5 but ranked too low).

These findings have practical implications. LLMs may be most useful as differential generators in the Rare tier, but should not be relied upon for Ultra-rare conditions without retrieval augmentation.